# Ingredient class, botanicals/herbs/traditional formulas,: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified standard for Ingredient class: botanicals/herbs/traditional formulas (Dietary Supplements (Human)), stated as sold. Every limit is read across from the strictest sovereign infant-and-child-food maximum level for the analyte and applied to all supplements as a voluntary child-protective over-compliance; the laxer adult supplement floor (sovereign supplement MLs, USP <2232>, Prop 65) is disclosed and never selected.

## 1 | How every number on this standard is set

HMTc Method v2.0 — child-protective over-compliance: strictest sovereign infant-and-child-food ML by lookup, applied to all supplements; adult floor (supplement ML / USP <2232> / Prop 65 MADL) disclosed, never selected

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 10 | human-grade read-across (Australia / New Zealand) |
| Cadmium (Cd) | Tier 1 | 5 | human-grade read-across (European Union) |

| | | | |
|---|---|---|---|
| Total arsenic (tAs) | Tier 2 | 50 | human-grade read-across (India) |
| Inorganic arsenic (iAs) | Tier 1 | 10 | human-grade read-across (European Union) |
| Mercury, total (tHg) | Tier 1 | 4 | human-grade read-across (Israel) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 100 | human-grade read-across (European Union) |
| Tin (Sn) | Tier 2 | 5,000 | human-grade read-across (India) |
| Aluminium (Al) | Tier 2 | 500 | human-grade read-across (Australia / New Zealand) |
| Total chromium (Cr) | Tier 2 | 1,000 | human-grade read-across (China) |
| Hexavalent chromium (Cr-VI) | Tier 2 | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 10 | Australia / New Zealand — infant formula products |
| Cadmium (Cd) | 5 | European Union — infant formulae, follow-on formulae, food for special medical purposes intended for infants and young children, and young-child formulae placed on the market as liquid and manufactured from cow's milk proteins or cow's milk protein hydrolysates |
| Total arsenic (tAs) | 50 | India — Infant Milk substitute and Infant foods |

| | | |
|---|---|---|
| Inorganic arsenic (iAs) | 10 | European Union — infant formulae, follow-on formulae, food for special medical purposes intended for infants and young children, and young-child formulae placed on the market as liquid |
| Mercury, total (tHg) | 4 | Israel — Infant and toddler formulas, ready-to-eat |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 100 | European Union — infant/young-child formula, liquid |
| Tin (Sn) | 5,000 | India — Infant Milk substitute and Infant foods |
| Aluminium (Al) | 500 | Australia / New Zealand — infant formula, follow-on formula, and special medical purpose product for infants other than preterm-infant products |
| Total chromium (Cr) | 1,000 | China — human-food analog |
| Hexavalent chromium (Cr-VI) | ≤ Cr | Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell. |

## 4 | Cite this standard

Pendergrass, K. "Ingredient class, botanicals/herbs/traditional formulas,." In Heavy Metal Tested & Certified (HMTc) Dietary Supplements (Human) standard, version 1.0 (2026-08-14). Institute of Contaminant Standards (ICS), 2026.

## References


[1] Australia / New Zealand — human-food maximum levels
Binds Pb, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] China — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Cd, iAs, Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] Israel — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[5] India — human-food maximum levels
Binds tAs, Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

## STATUS AND LICENCE

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## COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

## DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/supplements--botanicals-herbs. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/supplements--botanicals-herbs.